

Multiple Choice (10 marks)

1. Of the following problems concerning a given undirected graph G , which is currently known to be solvable in polynomial time?
 - (a) Finding a longest simple cycle in G
 - (b) Finding a shortest cycle in G
 - (c) Finding ALL spanning trees of G
 - (d) Finding a largest clique in G
2. In Python 3, which of the following function convert a string to an int in python?
 - (a) `int(x [,base])`
 - (b) `long(x [,base] )`
 - (c) `float(x)`
 - (d) `str(x)`
3. What is the output of the statement `"a" + "ab"` in Python 3?
 - (a) Error
 - (b) aab
 - (c) ab
 - (d) a ab
4. What is the output of `"abc"[::-1]` in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
5. In Python 3, `b = [11,13,15,17,19,21]`; `print(b[::2])` outputs what?
 - 19,21
 - 11,15
 - 11,15,19
 - 13,17,21

True / False (10 marks)

Answer True or False

6. True or False: Even after extensive testing, the program may still contain undetected bugs.
7. True or False: The value of the Python expression $4 + 3 \% 2$ is 6, since $3 \% 2$ equals 1 and is added to 4.
8. True or False: $O(n^2)$ is the largest asymptotic notation among $O(1)$, $O(n)$, $O(n^2)$, and $O(\log n)$.
9. True or False: The access matrix approach can accurately express complex protection requirements without limitations.
10. True or False: Python supports automatic garbage collection and can be integrated with languages like C, C++, and Java.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract all the pairs which are symmetric in the given tuple list.
12. Write a function to zip the two given tuples.

End of Paper